



# Metrics / validation evidence

Every number below is reproducible from the repo — and we state what we can't claim yet.

## 1 • Strategy — backtested on the venue's own data

| <b>12.2%</b>                  | <b>0.21%</b> | <b>4.5–22.6%</b>                | <b>~14.3%/yr</b>                                               |
|-------------------------------|--------------|---------------------------------|----------------------------------------------------------------|
| APY · 3.2y blended            | max drawdown | rolling-1y range (median 11.7%) | avg funding longs paid                                         |
| Dataset                       |              | <b>27,534 × 2</b>               | all hourly BTC+ETH prints since May 2023                       |
| Per-asset sleeve APY          |              | <b>12.3% / 12.1%</b>            | cBTC / cETH, independent sign guards                           |
| Time deployed / round trips   |              | <b>84% · 28–36</b>              | guard exits negative regimes; churn costs charged              |
| Leverage / capital efficiency |              | <b>5× · 83%</b>                 | liquidation-buffer bound $L \leq 1/(\text{maint}+\text{move})$ |

Methodology: non-lookahead (decide on trailing 3-day window, accrue the next print), both-legs trading costs on every open/unwind, realized funding only, no cherry-picked window — the range is every rolling year in the data. Today's compressed regime (~4.5%) is the bottom of the range, stated plainly. Reproduce: `python -m basisvault_engine.backtest`.

## 2 • Working software — real transactions, not mockups

| <b>6/6 steps</b>           | <b>7 + 38</b>               | <b>5-1-0-0</b>                | <b>24/7</b>      |
|----------------------------|-----------------------------|-------------------------------|------------------|
| lifecycle as real Daml txs | tests green (Daml + engine) | per-role ACS, ledger-enforced | live public demo |

The public demo executes deposit → open carry (at live Hyperliquid marks) → accrue funding → transfer → sign-guard unwind → redeem as actual Daml transactions on a Canton ledger via the JSON Ledger API v2 — real update ids shown in the UI. The privacy proof is Canton's own answer: after a full run the issuer's party sees 5 contracts, the observer 1, the holder 0, an outsider 0 — queried per-party from the ledger's active-contract set, not filtered by the UI.

## 3 • Provenance — the mechanism runs in production

BasisYield is not a hackathon sketch: basisyield.com runs this cash-and-carry engine 24/7 against live Hyperliquid data — real funding prints, oracle marks, maker/taker cost model, a layered killswitch, and paper-validated entries (e.g. short BTC perp + long spot at +11% funding APR, 5× perp, 83% capital efficiency). The Canton build ports its exact entry/exit rules and its honesty rules: three number types (projected / backtested / realized), never mixed.

## What we do NOT claim

No real-money track record yet — 12.2% is a backtest, and the production engine trades paper, not live capital. Funding is regime-dependent: compressed regimes degrade toward the collateral floor, not below it (the sign guard never holds through negative funding). In the demo, Hyperliquid order execution is attested on-ledger but not armed; the Canton ledger, workflow, privacy and data are real. Next-phase RWA-margin stacking (8.7% APY, 5y) is labelled a projection.